UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 011 (VoP) · Method of variation of parameters · compiled 16 October 2026

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1 Short notes

1 · The method: $y_p = uy_1 + vy_2$ with the Wronskian
Variation of parameters gives the PI of $y'' + py' + qy = X$ once the CF $c_1y_1 + c_2y_2$ is known. Make the coefficient of y'' 1. Wronskian $W = y_1y_2' - y_2y_1' \neq 0$. $u = -\int \frac{y_2X}{W}dx$, $v = \int \frac{y_1X}{W}dx$ (no constants), $y_p = uy_1 + vy_2$, $y = y_c + y_p$. Works for any X and, with a known CF, for variable coefficients. Demerits: needs the CF; integrals can be hard.

2 · Slide and lecture examples: xe^{-x}, $\tan 2x$, $\frac{2}{1+e^x}$, a Cauchy equation
$y'' - 2y' - 3y = xe^{-x}$: $W = -4e^{2x}$, $y_p = -\frac{e^{-x}}{16}(x + \frac{1}{4}) - \frac{x^2e^{-x}}{8}$. $(D^2 + 4)y = \tan 2x$: $W = 2$, $y_p = -\frac{1}{4}\cos 2x \log(\sec 2x + \tan 2x)$ (slide box drops $\cos 2x$). $(D^2 - 1)y = \frac{2}{1+e^x}$: $W = -2$, $y_p = e^x \log(1 + e^{-x}) - e^{-x}\log(1 + e^x) - 1$. Cauchy $x^2y'' - 3xy' + 3y = 12x^4$: in z, $W = -2e^{4z}$, $y = c_1x^3 + c_2x + 4x^4$.

3 · Trigonometric right-hand sides: $\sec ax$, $\csc x$, $e^x \tan x$, $e^{-x}\sec^3 x$
Roots $\alpha \pm i\beta$, $X = e^{\alpha x}g$: $W = \beta e^{2\alpha x}$, $u = -\frac{1}{\beta}\int g \sin\beta x$, $v = \frac{1}{\beta}\int g\cos\beta x$. $\sec ax$: $\frac{\cos ax}{a^2}\log\cos ax + \frac{x}{a}\sin ax$. $\csc x$: $-x\cos x + \sin x \log\sin x$. $e^x\tan x$ (ESA): $-e^x\cos x\log(\sec x + \tan x)$. $e^{-x}\sec^3 x$ (ESA): $\frac{1}{2}e^{-x}\sin x\tan x$.

4 · Repeated roots with $X = \frac{e^{mx}}{x^k}$, and the leading-coefficient trap
Repeated root m: $y_1 = e^{mx}$, $y_2 = xe^{mx}$, $W = e^{2mx}$. $X = \frac{e^{mx}}{x}$ (ESA ×3): $u = -x$, $v = \log x$, $y = (c_1 + c_2x)e^{mx} + xe^{mx}\log x$. $X = \frac{e^{-x}}{x^2}$: $y_p = -e^{-x}\log x$. Leading coefficient 2: divide first (CW P27).

2 Keyword index

20 terms; the number in brackets is the topic.

$\frac{e^{mx}}{x}$ (4). ESA ×3: $y_p = xe^{mx}\log x$.

$\int \sec$ (2). $\log(\sec x + \tan x)$.

$\int \tan$, $\int \cot$ (3). $\log\sec x$, $\log\sin x$.

$e^x \tan x$ (3). ESA Jan-24 s1 3a: $y_p = -e^x \cos x \log(\sec x + \tan x)$.

$e^{-x}\sec^3 x$ (3). ESA Jan-24 s2 3a: $y_p = \frac{1}{2}e^{-x}\sin x \tan x$.

u (1). $-\int \frac{y_2 X}{W}dx$.

v (1). $\int \frac{y_1 X}{W}dx$.

Absorbing (1). Dropping multiples of y_1, y_2 from y_p into c_1, c_2.

Cauchy's equation (2). $x^2y'' + \ldots$: CF via $x = e^z$, then VoP in z.

Complementary function (1). $c_1y_1 + c_2y_2$; needed first.

Complex roots (3). $\alpha \pm i\beta$: $W = \beta e^{2\alpha x}$.

Cramer's rule (1). Solves $u'y_1 + v'y_2 = 0$, $u'y_1' + v'y_2' = X$.

Integration by parts (2). $\int xe^{-4x}dx$ in slide 011V·9.

Linear independence (1). $c_1y_1 + c_2y_2 \equiv 0 \Rightarrow c_1 = c_2 = 0$; $W \neq 0$.

Partial fractions (2). $\frac{1}{e^x(1+e^x)} = \frac{1}{e^x} - \frac{1}{1+e^x}$.

Particular integral (1). $y_p = uy_1 + vy_2$.

Repeated root (4). $y_2 = xe^{mx}$, $W = e^{2mx}$.

Standard form (1). Coefficient of y'' equal to 1.

Variation of parameters (1). Replace c_1, c_2 by functions $u(x)$, $v(x)$ to get the PI.

Wronskian (1). $W = y_1y_2' - y_2y_1' \neq 0$.

3 Rapid recall

1	What does variation of parameters find?	Only the PI; the CF must be known first.
2	Standard form required?	$y'' + py' + qy = X$: coefficient of y'' equal to 1.
3	Wronskian?	$W = y_1 y_2' - y_2 y_1'$.
4	u and v?	$u = -\int \frac{y_2 X}{W} dx$, $v = \int \frac{y_1 X}{W} dx$.
5	PI?	$y_p = uy_1 + vy_2$.
6	Why no constants of integration?	They would add ky_1, ky_2, already in the CF.
7	Three steps?	CF (y_1, y_2) → Wronskian → u, v, PI.
8	W for e^{3x}, e^{-x}?	$-4e^{2x}$.
9	W for $\cos 2x$, $\sin 2x$?	2.
10	PI of $(D^2+4)y = \tan 2x$?	$-\frac{1}{4}\cos 2x \log(\sec 2x + \tan 2x)$ (slide box drops $\cos 2x$).
11	PI of $(D^2-1)y = \frac{2}{1+e^x}$?	$e^x \log(1+e^{-x}) - e^{-x}\log(1+e^x) - 1$.
12	W for complex roots $\alpha \pm i\beta$?	$\beta e^{2\alpha x}$.
13	ESA: $(D^2 - 2D + 2)y = e^x \tan x$?	$y = e^x(c_1 \cos x + c_2 \sin x) - e^x \cos x \log(\sec x + \tan x)$.
14	ESA: $y'' + 2y' + 2y = e^{-x}\sec^3 x$?	$y = e^{-x}(c_1 \cos x + c_2 \sin x) + \frac{1}{2}e^{-x}\sin x \tan x$.
15	ESA: $y'' + 6y' + 9y = \frac{e^{-3x}}{x}$?	$y = (c_1 + c_2 x)e^{-3x} + xe^{-3x}\log x$.
16	ESA: $y'' - 2y' + y = \frac{e^x}{x}$?	$y = (c_1 + c_2 x)e^x + xe^x \log x$.
17	CW P27 first step?	Divide by 2: $X = \frac{25}{2}e^{-x}$.
18	Cauchy $x^2y'' - 3xy' + 3y = 12x^4$ by VoP?	In $z = \log x$: $W = -2e^{4z}$, $y = c_1 x^3 + c_2 x + 4x^4$.

Mnemonic (study aid). *"u is minus, and takes the other one."* $u = -\int \frac{y_2 X}{W}$, $v = +\int \frac{y_1 X}{W}$: each parameter integrates the *other* solution (study aid).

4 Last 10 minutes before the exam

Top facts

1. VoP gives the PI only; find the CF first; make the coefficient of y'' equal to 1.
2. $W = y_1y_2' - y_2y_1'$; $u = -\int \frac{y_2X}{W}dx$, $v = \int \frac{y_1X}{W}dx$, no constants; $y_p = uy_1 + vy_2$; $y = y_c + y_p$.
3. Complex roots $\alpha \pm i\beta$: $W = \beta e^{2\alpha x}$; repeated root m: $W = e^{2mx}$.
4. ESA ×3: $(D-m)^2y = \frac{e^{mx}}{x} \rightarrow y = (c_1 + c_2x)e^{mx} + xe^{mx}\log x$ (absorb $-xe^{mx}$).
5. ESA: $e^x \tan x$ over $D^2 - 2D + 2 \rightarrow -e^x \cos x \log(\sec x + \tan x)$; $e^{-x}\sec^3 x$ over $D^2 + 2D + 2 \rightarrow \frac{1}{2}e^{-x}\sin x \tan x$.
6. $\tan 2x$ over $D^2 + 4 \rightarrow -\frac{1}{4}\cos 2x \log(\sec 2x + \tan 2x)$ (the slide's box drops $\cos 2x$).
7. Integrals: $\int \tan = \log\sec$, $\int \cot = \log\sin$, $\int \sec = \log(\sec + \tan)$, $\frac{\sin^2}{\cos} = \sec - \cos$.

Top formulas

E1.2 · Wronskian ESA ★★★

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_2y_1' \neq 0$$

GIVES The denominator of both integrals.

APPLIES y_1, y_2 linearly independent.

NOTE Order matters: $W(y_2, y_1) = -W(y_1, y_2)$, but the PI is unchanged (T011Vb 00:09:20).

S011V·5 · S011V·8 · T011V 00:02:35 · T011V 00:10:22 · ESA Jul-23 3c

E1.3 · The parameters u and v ESA ★★★

$$u = -\int \frac{y_2\,X}{W}\,dx, \qquad v = \int \frac{y_1\,X}{W}\,dx$$

GIVES The two varying parameters.

APPLIES Every VoP problem.

NOTE Minus sign on u (with y_2); no constants of integration.

S011V·5 · T011V 00:03:26 · T011V 00:04:18 · ESA Jul-23 3c · ESA May-24 3c · ESA Jan-24 s1 3a · ESA Jan-24 s2 3a

E1.4 · The particular integral and the general solution ESA ★★★

$$y_p = u\,y_1 + v\,y_2, \qquad y = c_1y_1 + c_2y_2 + y_p$$

GIVES The PI and, with the CF, the general solution.

APPLIES After E1.3.

NOTE Terms of y_p that are multiples of y_1 or y_2 may be absorbed into c_1, c_2.

S011V·5 · S011V·10 · S011V·17 · T011V 00:14:11 · T011Vb 00:16:16

E3.1 · Complex roots $\alpha \pm i\beta$ with $X = e^{\alpha x} g(x)$ ESA ★★★

$$y_1 = e^{\alpha x}\cos\beta x, \quad y_2 = e^{\alpha x}\sin\beta x, \quad W = \beta e^{2\alpha x}$$
$$u = -\frac{1}{\beta}\int g\sin\beta x\,dx, \qquad v = \frac{1}{\beta}\int g\cos\beta x\,dx$$

GIVES The integrals with the exponentials already cancelled.

APPLIES CW P23, P25, P26, P28; ESA Jan-24 s1 3a, s2 3a.

NOTE $\alpha = 0$ for $D^2 + a^2$ ($W = a$); $D^2 - 2D + 2$: $1 \pm i$; $D^2 + 2D + 2$: $-1 \pm i$.

ADDED FOR CLARITY S011V·11 · CW U2 P23 · CW U2 P25 · ESA Jan-24 s1 3a · ESA Jan-24 s2 3a

E4.2 · Results: $\frac{e^{mx}}{x}$ and $\frac{e^{mx}}{x^2}$ ESA ★★★

$$(D-m)^2 y = \frac{e^{mx}}{x}: \quad y = (c_1 + c_2 x)e^{mx} + xe^{mx}\log x$$
$$(D-m)^2 y = \frac{e^{mx}}{x^2}: \quad y_p = -e^{mx}\log x$$

GIVES The ESA answers with $m = -3$ (Jul-23, Jun-24) and $m = 1$ (May-24); CW P24 with $m = -1$.

APPLIES After absorbing CF terms.

NOTE Raw $uy_1 + vy_2$ has an extra $-xe^{mx}$ (resp. $-e^{mx}$): a CF term.

ESA Jul-23 3c · ESA Jun-24 3a · ESA May-24 3c · CW U2 P24